

Supplementary Information

Host-Conditioned Spatial Structure of Canopy Mortality in Finland’s Boreal Forests

immediate

This Supplement records the software environment and random seeds (§1), the shape-feature correlation structure (§2) and univariate feature AUCs (§3) that underpin the morphological-gradient framing, the per-figure provenance trail (§4), and a set of supporting figures relocated from the main text (§5). It accompanies the main text and uses the same cluster table (14,582 clusters; 3,422 elongated/dispersed “Type0” and 11,160 compact/aggregated “Type1”).

1 Software environment and random seeds

Component	Version / value
Python	3.9.6
numpy / scipy	2.0.2 / 1.13.1
pandas / scikit-learn	2.3.3 / 1.6.1
geopandas / shapely / pyproj	1.0.1 / 2.0.7 / 3.6.1
rasterio / pointpats	1.4.3 / 2.3.0
matplotlib	3.9.4
R	4.6.1
spatstat (.geom/.explore)	3.6.1 (3.8.1)
Covariance-preserving null seed	42
Discreteness tests (GMM/PCA) seed	42
Tree-level subsample seed	42
spatstat envelopes	39 sims/block
DBSCAN eps (clusters / blocks)	20 m / 15 km
DBSCAN min_samples	1 (deterministic)
Host/species raster vintage	MS-NFI vmi1x_1721 (2021)

Table S1: Software versions and seeds for full reproducibility. All stochastic steps are seeded; DBSCAN is deterministic at `min_samples = 1`.

2 Shape-feature correlation structure

The morphological features are strongly intercorrelated (Figure S1). Aspect ratio and elongation are effectively the same axis ($\rho = 1.00$); compactness is its near-perfect inverse ($\rho = -0.85$); area, perimeter, core:edge ratio and point count form a tight size block ($\rho \geq 0.71$). The three features used to define the $k = 2$ partition (fractal dimension, Clark–Evans, aspect ratio; red boxes) therefore span only two largely independent directions, an elongation axis and a size/density axis, which is why a single principal component captures 60% of the variance (main text, §Morphological Gradient) and why the partition is best read as a cut through a continuum rather than two natural classes.

3 Univariate feature separability

Table S2 reports the area under the ROC curve for each feature taken alone as a classifier of the two morphological endpoints (direction: “+” = larger values associate with the compact endpoint). The high AUCs of compactness (0.961), aspect ratio (0.945) and elongation (0.945) are *expected and partly circular*: the partition is defined on aspect ratio and fractal dimension, so any feature coupled to those axes separates the endpoints almost perfectly by construction. These values quantify the geometric redundancy among descriptors discussed in the main text;

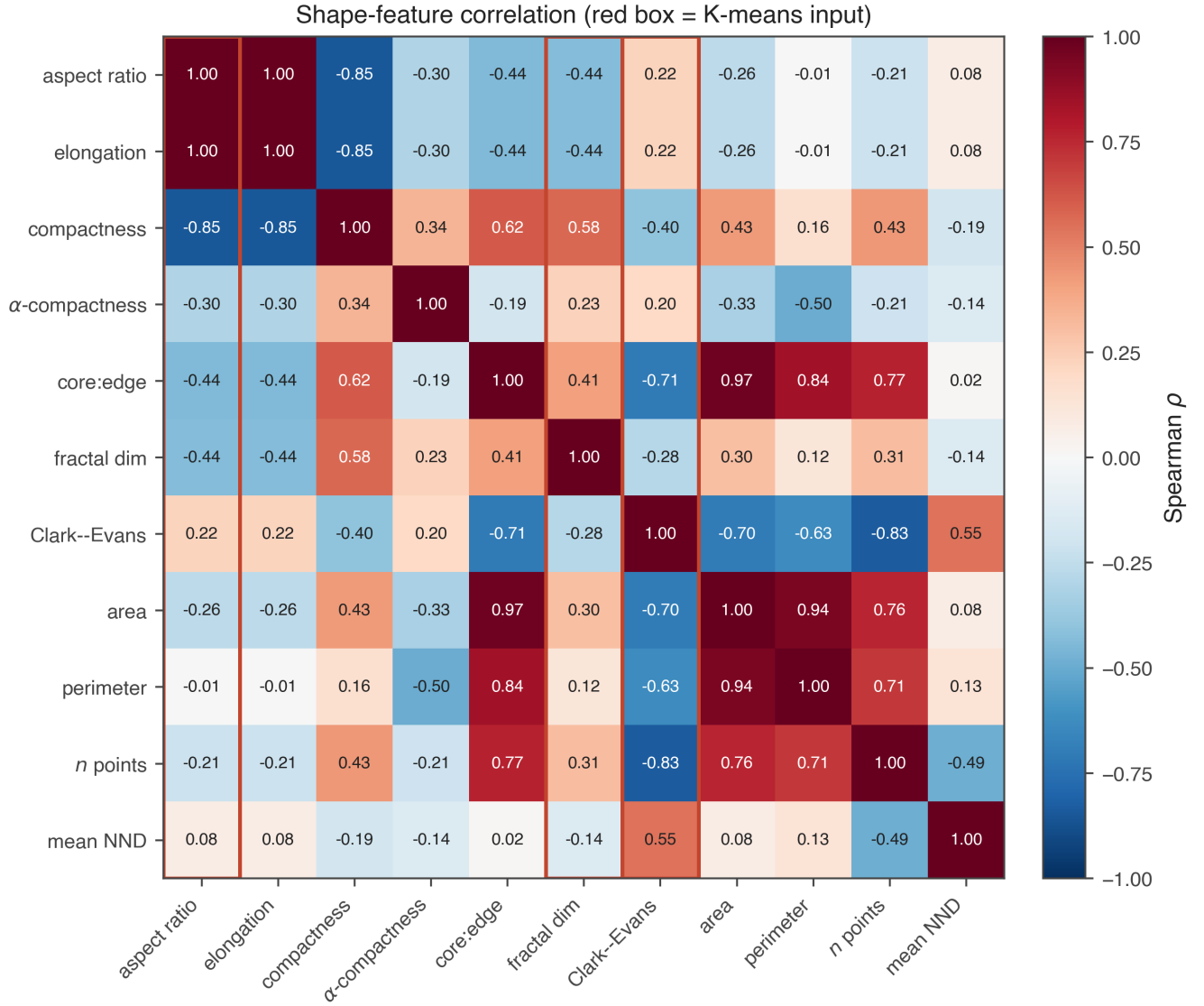


Figure S1: Spearman correlation among the eleven cluster-level shape and point-pattern features. Red boxes mark the three features input to $k = 2$ K-means.

they are not independent evidence for two classes. The one shape-orthogonal quantity examined in the main text, landscape connectivity, adds no separating signal once size is accounted for.

4 Per-figure provenance

Every main-text figure is regenerated by a script in the public repository (<https://github.com/aniskhan25/TreeMortSpatial>, analysis/); the `restyle_*` scripts apply the shared house style (`analysis/figstyle.py`) to notebook-produced figures.

5 Supporting figures relocated from the main text

These descriptive and supporting figures are referenced from the main text. They were moved here to keep the main figures focused on the three contributions; the analyses they depict are unchanged.

Feature	AUC	Dir. (compact)
compactness	0.961	+
aspect ratio [†]	0.945	−
elongation	0.945	−
fractal dim [†]	0.938	+
core:edge	0.837	+
Clark–Evans [†]	0.760	−
area	0.733	+
n points	0.726	+
α -compactness	0.662	+
mean NND	0.622	−
perimeter	0.577	+

Table S2: Univariate ROC-AUC for separating the elongated (Type0) from the compact (Type1) endpoint. [†] = feature used to define the $k = 2$ partition. Direction “+” = larger values associate with the compact endpoint. High AUCs reflect geometric redundancy, not independent validation.

Figure	Generating script	Primary input
fig2_histograms	make_histograms_figure.py	cluster table
fig4_scatters	restyle_charts.py	cluster table
fig6_typology_map	restyle_maps.py	cluster table
fig8_local_morans	restyle_maps.py	gate_moran*
fig_shape_space	restyle_charts.py	cluster table
fig_size_dist	restyle_charts.py	cluster table
fig_orientation_rose	restyle_charts.py	cluster table
fig_moran_heatmap	restyle_charts.py	gate_moran*
fig_silhouette_permtest	make_permtest_figure.py	discreteness_tests.py
fig_dbscan_sensitivity	make_gate_figure.py	gate1_eps_stability
fig_anisotropy	make_anisotropy_figure.py	gate3_anisotropy
fig_shap_heldout	restyle_shap.py	held-out RF
fig_contagion_a/b	restyle_maps.py	cluster table
fig_sampling_extent	scope_experiments.py	cluster table
fig_host_conditioned	make_hostcond_figure.py	host_pointprocess.py
fig_hostcond_rigor	make_hostcond_rigor_figure.py	linhom*.csv
fig_species_validation	species_validation.py	MS-NFI species rasters
fig_si_corr (this SI)	make_si_tables.py	cluster table

Table S3: Per-figure provenance. “cluster table” = data/clusters_typed.csv; gate/host artifacts are written to analysis/out/.

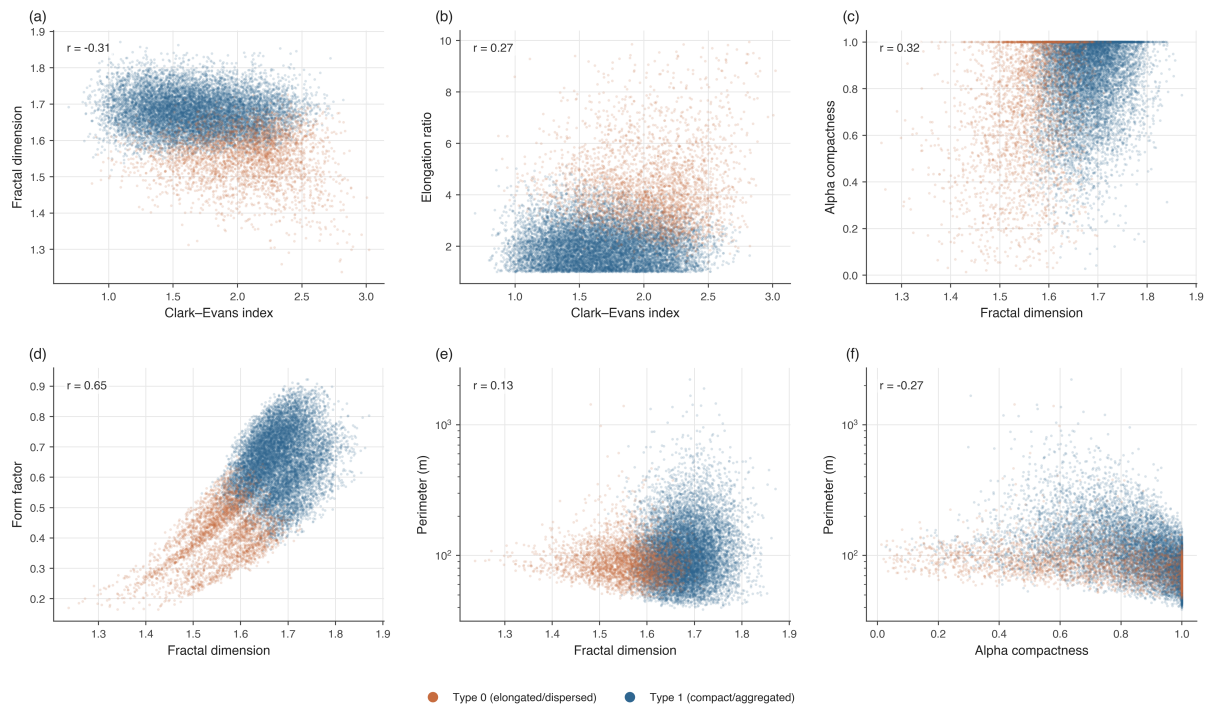


Figure S2: Pairwise scatter plots of selected spatial metrics, coloured by morphological endpoint.

Gradient endpoints — study region ($n=1,585$)

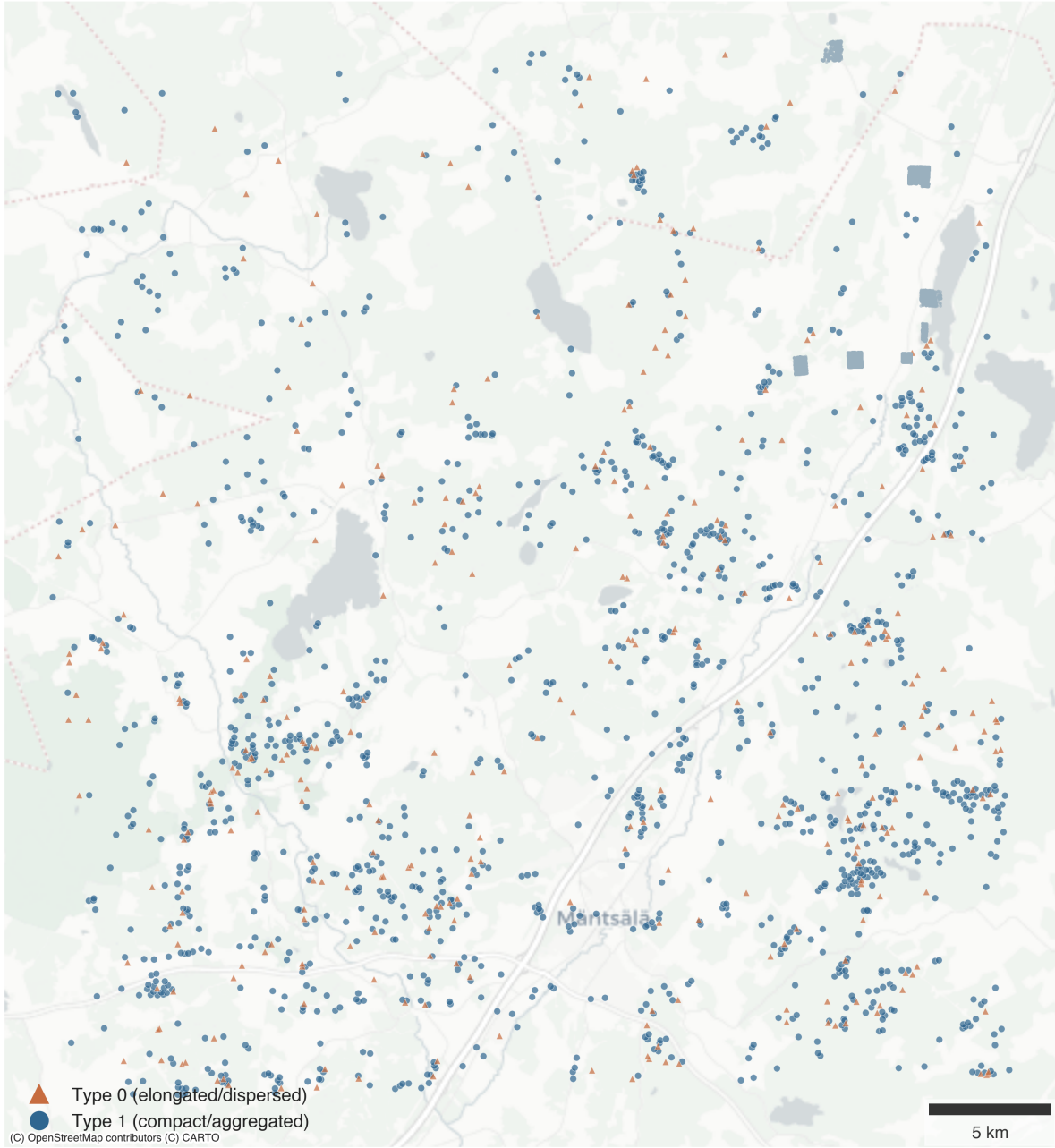


Figure S3: Spatial distribution of the two gradient endpoints within a southern display window (one tile, 10.9% of clusters, shown for legibility). Blue circles: Type 1 (compact/aggregated); brown triangles: Type 0 (elongated/dispersed). Inset: all 14,582 clusters sampled across Finland for geographic context.

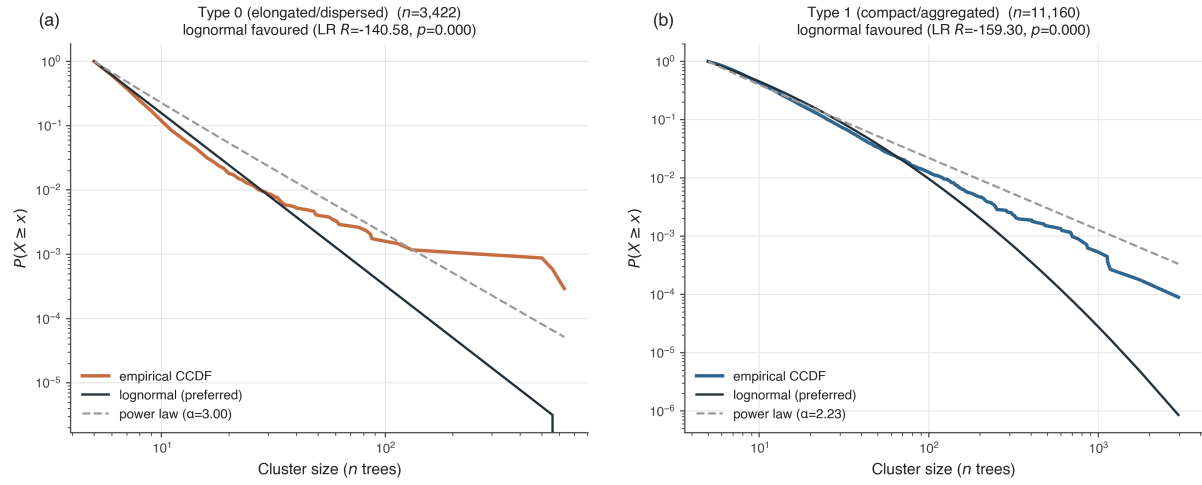


Figure S4: Complementary cumulative distribution functions (CCDF) of cluster size (n) per endpoint. Both are better described by a lognormal than a power law; Type 1 has the heavier upper tail, consistent with its larger mean cluster area.

Local Moran's I — fractal-dimension hot/cold spots

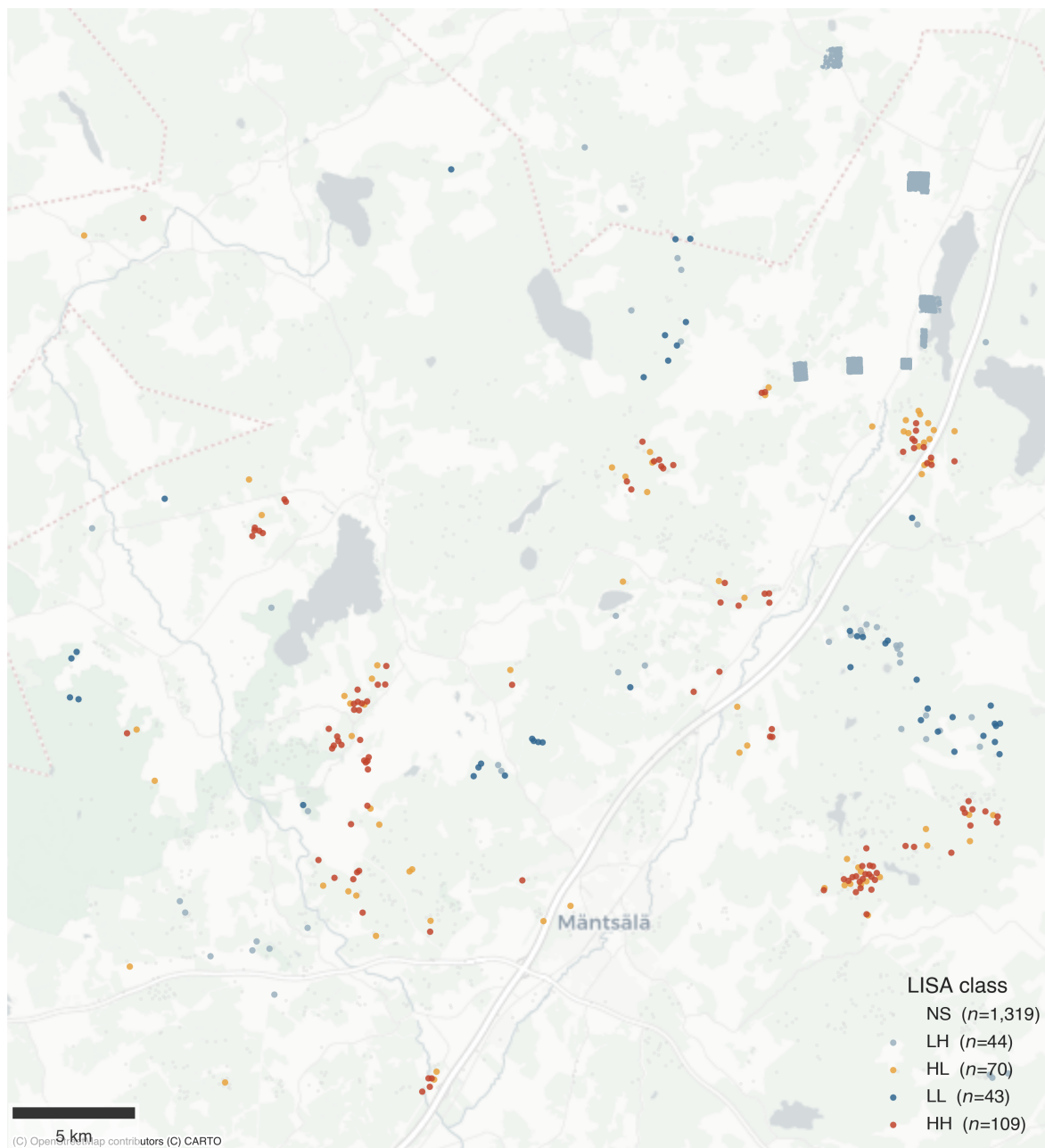
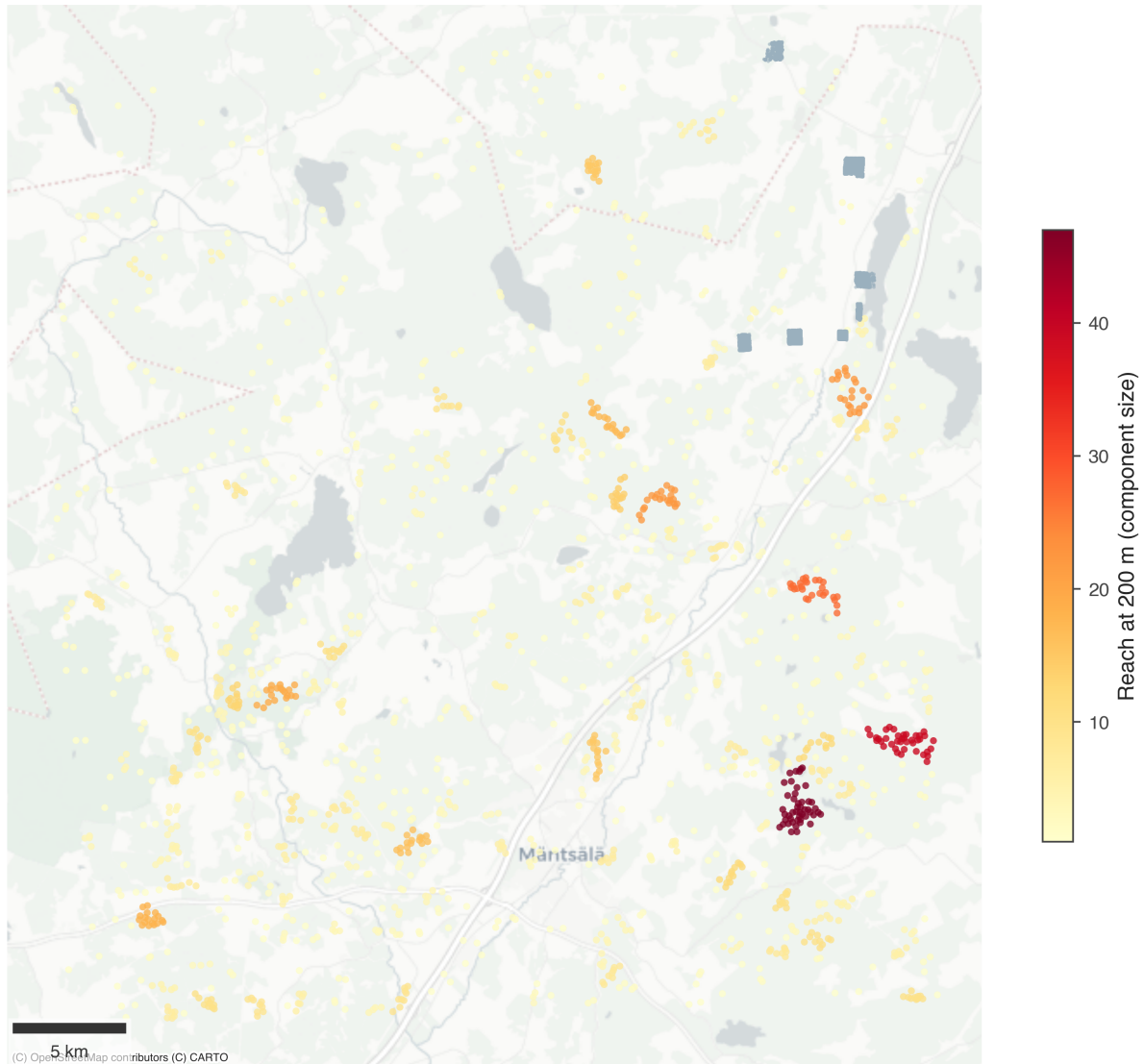


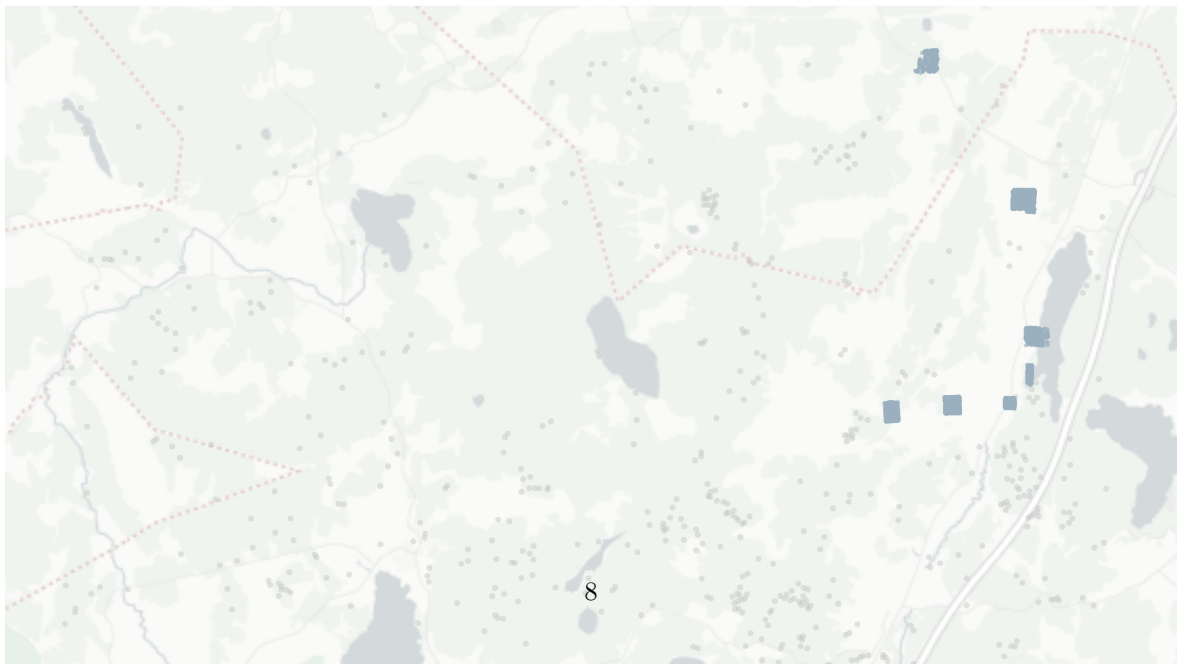
Figure S5: LISA hotspot map for fractal dimension in the southern display window. Labels computed on the full survey dataset. Red: high-high clusters; blue: low-low clusters. Inset: full-survey cluster distribution for geographic context.

Connectivity reach at 200 m



(a) Connectivity reach at 200 m (component size). Inset: full-survey cluster distribution.

High-betweenness bridge clusters
(geometric articulation points; no temporal validation)



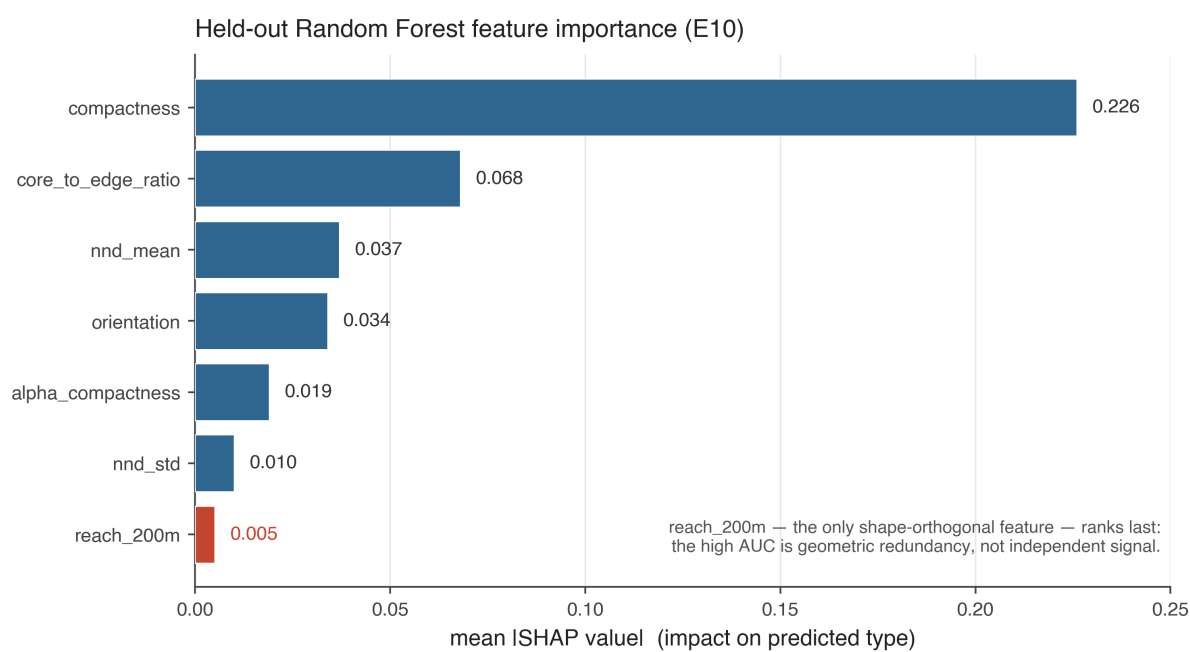


Figure S7: Mean |SHAP| feature importance for the held-out Random Forest. Compactness dominates; reach_200m (the only shape-orthogonal feature, highlighted) ranks last, so the high AUC reflects geometric redundancy among shape descriptors rather than independent signal.